AGRO

Autonomous Growth and Resources Observer



What is AGRO?

AGRO is a robot that drives through a garden, detects the crops it sees, and records the count on a dashboard. A webcam feeds a YOLO11n model running on a Raspberry Pi 4. An Arduino Uno and an L298N driver handle the motors, the soil probe, and the water pump, and a soil moisture sensor reads the soil. I built it as an Ackermann steering chassis that I reused from an old RC car.

Where it started

AGRO began as our innovative project entry in the agriculture category of the DLSMHSI Robosports Tournament. I built V1 with my senior, Princess Frias, the former president of Animo SPEAR, and my batchmate, Mikoh Filoteo. I was in-charge of the hardware and software while they focused on research and experiments. After the tournament I kept improving it, which became V2.

What was the Version 1?



Figure 1. AGRO Version 1

V1 used RFID and a basic soil sensor, and recorded inventory in a Google Sheet through an ESP8266. It was costly and unsafe to handle, and the soil probe had to be dangled by hand.

The 3D Model of the AGRO Version 2.

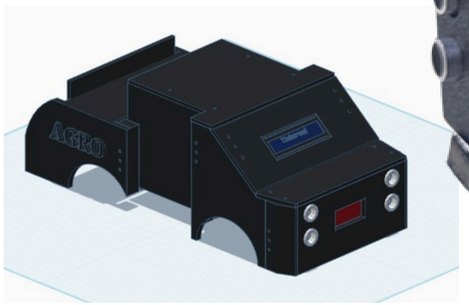


Figure 2. AGRO Version 2 3D Model



Figure 3. Webcam View

Weak obstacle avoidance on my first RC chassis traced back to its Ackermann steering geometry.



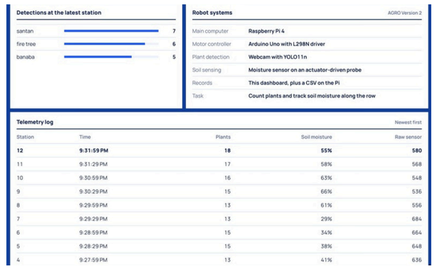


Figure 4 and 5. The dashboard



Figure 6. AGRO V2 testing

What changed in Version 2

V2 replaces RFID with a webcam and YOLO11n on the Raspberry Pi, so AGRO identifies crops by sight and leaves wilting plants out of the count. If YOLO finds nothing, a TFLite SSD model checks the same frame. An actuator now lowers the soil probe and pushes it into the soil, so nobody has to hold it. The Google Sheet and ESP8266 became a dashboard. I tested it in our school's botanical garden.

Where it all started



Figure 7. AGRO at the Robosports tournament

Our V1 at the Robosports tournament, built with Princess Frias and Mikoh Filoteo.

AGRO: HOW IT DECIDES

Seeing the crops

A webcam feeds YOLO11n on the Raspberry Pi, trained on my own plant photos and run at 320 px so it stays fast. If YOLO finds nothing in a frame, a TFLite SSD model checks the same frame. In every 10-second window, AGRO keeps the highest count of each crop it saw at once, saves it to a CSV file with the sensor readings, and sends it to the dashboard.

Reading and watering the soil

Every 10 minutes the Pi asks the Arduino for a soil reading. The robot stops, lowers the probe, averages 15 samples, and lifts it again. A DHT22 adds temperature and humidity, and a level sensor guards the tank. If the soil is under 35% and the tank is not nearly empty, the pump runs for 4 seconds, then rests for 10 minutes. An ultrasonic sensor backs the robot away from anything closer than 25 cm.

Forecasting the next watering

After watering, soil dries toward a level where it settles. AGRO fits that curve to each drying episode and estimates the minutes left until the soil hits the watering threshold. Heat and humidity change how fast soil dries, so after four finished episodes it also learns that link, which gives a new episode an early estimate. A jump of more than 8 points marks a new episode, because it means watering or rain.

The drying model

$$\theta(t) = \theta_{eq} + (\theta_0 - \theta_{eq}) \cdot e^{(-t/\tau)}$$

θ is soil moisture in %, θ_{eq} is the level the soil settles at, and τ is how slowly it dries. The time until the 35% watering threshold, from the current reading θ_{now}, is:

$$t = \tau \cdot \ln[(\theta_{now} - \theta_{eq}) / (35 - \theta_{eq})]$$

To find τ and θ_{eq}, the Pi spaces the readings evenly in time and fits each reading against the next with a straight line, $\theta[k+1] = a \cdot \theta[k] + b$. Then $\tau = -\Delta t / \ln a$ and $\theta_{eq} = b / (1 - a)$. A fit only counts if $0 < a < 1$, which means the readings really form a decaying curve.



AGRO

Autonomous Growth and Resources Observer: Development and Testing Photos



